

1. GitHub Repository Link:

https://github.com/loannaGiag/seip_assignment_1_2026

2. CI/CD:

The screenshot shows the GitHub Actions page for the repository 'seip_assignment_1_2026'. The left sidebar contains a list of actions: 'All workflows', 'CI/CD Pipeline', 'Management', 'Caches', 'Attestations', 'Runners', 'Usage metrics', and 'Performance metrics'. The main area displays 'All workflows' with a search bar and a 'Filter workflow runs' input. Below this, there is a section for '10 workflow runs' with a table showing the status of various workflows. The first workflow is 'final', which is in a 'Running' state. The table also shows the commit hash 'adf85b5' and the actor 'loannaGiag'.

Workflow	Event	Status	Branch	Actor
final	main	Running	main	loannaGiag

3. Cluster State:

```
PS C:\Users\ioann\Πανεπιστήμιο\6οΕξάμηνο\seip_assignment_1_2026> kubectl get all -n default
NAME                                READY    STATUS    RESTARTS   AGE
pod/echo-api-deployment-5754c4c857-6j4pk  1/1      Running   1 (7m57s ago)  14m
pod/echo-api-deployment-5754c4c857-m65xt  1/1      Running   1 (7m57s ago)  15m
pod/echo-api-deployment-5754c4c857-rvm28  1/1      Running   1 (7m57s ago)  15m

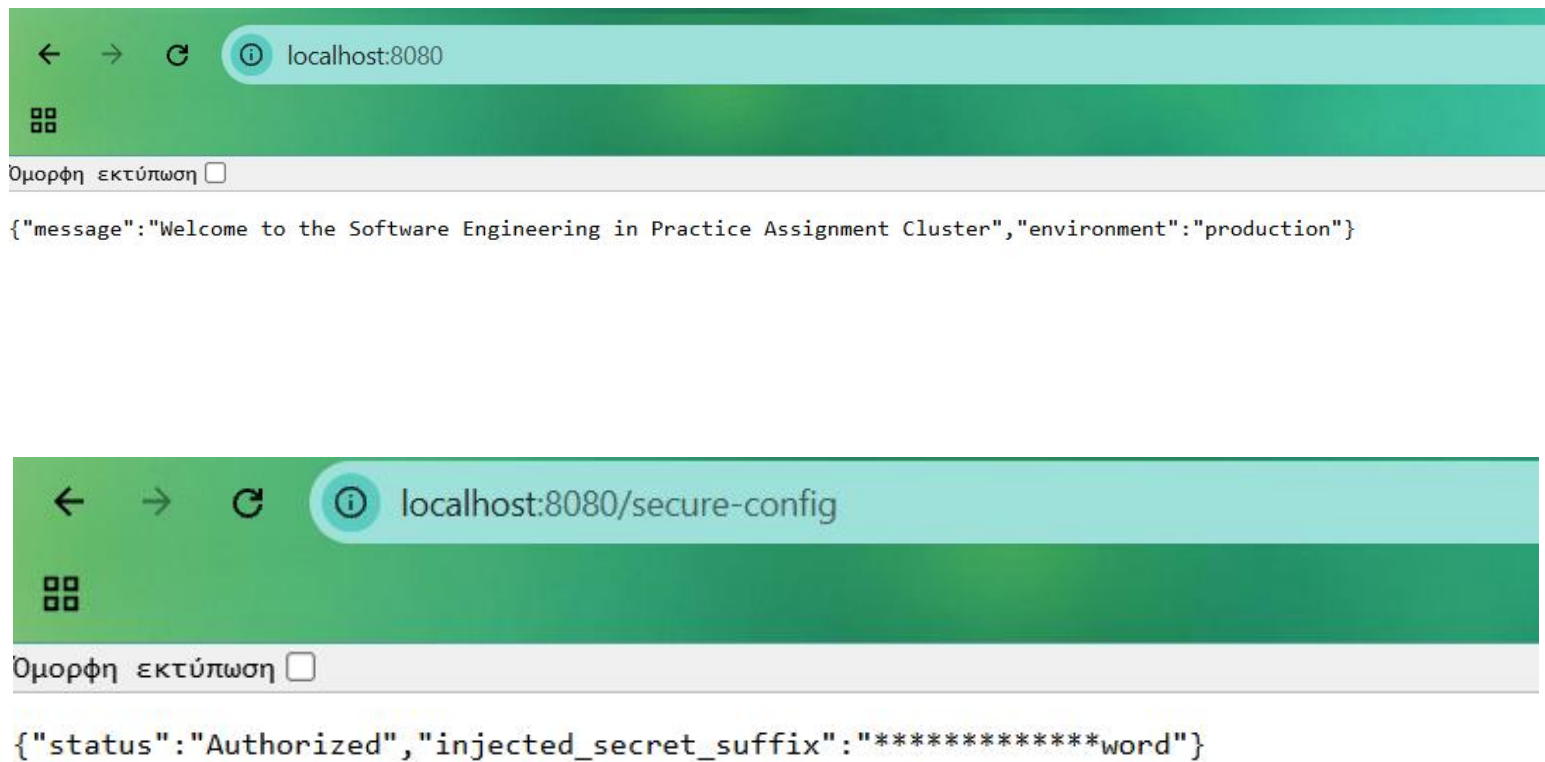
NAME                                TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/echo-api-service            ClusterIP     10.104.202.77 <none>         80/TCP     44m
service/kubernetes                  ClusterIP     10.96.0.1     <none>         443/TCP    46m

NAME                                READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/echo-api-deployment  3/3      3              3            44m

NAME                                DESIRED    CURRENT    READY    AGE
replicaset.apps/echo-api-deployment-5754c4c857  3          3          3        15m
replicaset.apps/echo-api-deployment-74dc957ddf  0          0          0        44m
PS C:\Users\ioann\Πανεπιστήμιο\6οΕξάμηνο\seip_assignment_1_2026> kubectl get configmap,secret
NAME                                DATA    AGE
configmap/echo-api-config           2        44m
configmap/kube-root-ca.crt          1        46m

NAME                                TYPE      DATA    AGE
secret/echo-api-secret              Opaque    1        44m
PS C:\Users\ioann\Πανεπιστήμιο\6οΕξάμηνο\seip_assignment_1_2026> |
```

4. Application Verification:



5. AI Reflection & Future Outlook:

During the assignment I used Generative AI tools for helping me understand the concepts and the tools we had to use for the assignment and how they work. It also helped me understanding the structure of the files and in troubleshooting. I found AI very useful in helping me understand what each section does and why it is needed.

On the other hand, I experienced some issues with the AI when I encountered problems related to Kubernetes pod failures, image updates, application deployment and when I had some issues in setting the tools needed on my laptop. In those situations, AI didn't always provide the correct solution immediately, so it required several rounds of discussion and testing before I actually solved those errors.

If I had more time to improve this project, I would try to make it more suitable for a real production environment by using an Ingress Controller, instead of port-forwarding, so the application can be accessed in a more proper and stable way. Also, I would add a GitOps workflow using ArgoCD so that deployments are handled automatically from the Git repository without manual commands. Furthermore, I would include monitoring tools like Prometheus and Grafana to better understand how the system is performing and to track the health of the application and finally, I would add security scanning in the CI pipeline to check Docker images for vulnerabilities before they are deployed.